Divisor of Algebraic Curves (Foundations Ver.)

Tugas Seminar I version
MA4091 Seminar Matematika I

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Table of Contents

1 Introduction

2 Noetherian, Zariski and Nullstellesatz

3 Affine Varieties

4 Sheaf & Affine Scheme

5 Closing & References

Description of this Presentation

- The Topic of this Final Paper is "Divisors of Algebraic Curves".
- The foundations of this topic are Classical Algebraic Geometry, Scheme Theory and Commutative Algebra.
- While the key machinaries of the topic are Sheaf Theory, Invertible Sheaves, and Weil/Cartier Divisors.
- Hence in this presentation, we shall present some foundational topics.

Motivation

- Given a polynomial $p(x)$, we wished to understand the geometry of the solutions of $p(x) = 0$.

Outline of this section

1. Noetherian Rings
2. Zariski Topology
3. Nullstellensatz Theorem

Noetherian Rings

Definition 1 (Noetherian Ring)

A ring A is *Noetherian* if the following equivalent conditions hold:

(i) Every ideal $I \subset A$ is finitely generated; that is, for every ideal $I \subset A$, there exist $f_1, \ldots, f_k \in I$ such that $I = (f_1, \ldots, f_k)$.

(ii) Every ascending chain

$$I_1 \subset I_2 \subset \cdots \subset I_m \subset \ldots$$

of ideals of A terminates, that is, the chain is eventually stationary, with $I_N = I_{N+1} = \ldots$ (the ascending chain condition, or a.c.c.).

(iii) Every nonempty set of ideals of A has a maximal element.

- For example, $\mathbb{Z}$ and $k[x]$ are Noetherian.
- The intuition of Noetherian Ring is we want to work in ring that has finite generators.

Proposition 2

(i) Suppose that A is Noetherian, and $I \subset A$ an ideal; then the quotient ring $B = A/I$ is Noetherian.
(ii) Let A be a Noetherian integral domain, and $A \subset K$ its field of fractions; let $\emptyset \neq S \subset A$ be a subset, and set

$$B = A[S^{-1}] = \left\{ \frac{a}{b} \in K \,\middle|\, a \in A,\ b = 1 \text{ or a product of elements of } S \right\}.$$

Then B is again Noetherian.

Theorem 3 (Hilbert Basis Theorem)

For a ring A,

$$A \text{ Noetherian} \Rightarrow A[X] \text{ Noetherian}.$$

Corollary 4

For k a field, a finitely generated k-algebra is Noetherian.

Algebraic set & Zariski

Definition 5

A subset $X \subset \mathbb{A}_k^n$ is an *algebraic set* if $X = V(I)$ for some ideal I. Notice that by Corollary 3.3, I is finitely generated. If $I = (f_1, \ldots, f_r)$ then clearly

$$V(I) = \{P \in \mathbb{A}_k^n \mid f_i(P) = 0 \text{ for } i = 1, \ldots, r\},$$

so that an algebraic set is just a locus of points satisfying a finite number of polynomial equations.

Zariski Topology

Proposition 6 (Proposition-Definition)

The correspondence V satisfies the following formal properties:

(i) $V(0) = \mathbb{A}_k^n; \quad V(A) = \emptyset;$

(ii) $I \subset J \implies V(J) \subset V(I);$

(iii) $V(I_1 \cap I_2) = V(I_1) \cup V(I_2);$

(iv) $V\left(\sum_i I_i\right) = \bigcap_i V(I_i).$

Hence the algebraic subsets of $\mathbb{A}_k^n$ form the closed sets of a topology on $\mathbb{A}_k^n$, the *Zariski topology*.

V-I Correspondence

$$\{\text{ideals } J \subset A\} \longleftrightarrow \{\text{subsets } X \subset \mathbb{A}_k^n\}$$
$$J \mapsto V(J) = \{P \in \mathbb{A}_k^n \mid f(P) = 0 \text{ for all } f \in J\}.$$

As a kind of inverse to V there is a correspondence

$$\{\text{ideals } J \subset A\} \longleftarrow \{\text{subsets } X \subset \mathbb{A}_k^n\}$$
$$I(X) = \{f \in k[X_1, \ldots, X_n] \mid f(P) = 0 \text{ for all } P \in X\}. \hookleftarrow X$$

That is, I takes a subset X to the ideal of functions vanishing on it.

Nullstellensatz

Polynomial Maps

Definition 7

Polynomial Maps Define $f : V \to W$ as polynomial maps, given by m polynomial functions $f_1, \ldots, f_m \in k[V]$,, so that for $P \in V$,

$$f(P) = \big(f_1(P), \ldots, f_m(P)\big) \in W.$$

- Keep in mind that if we have composite of f with any polynomial function $g \in k[W]$ is again polynomial function on V.
- We can see that polynomial maps evaluate each coordinates by its polynomial functions, hence we have such proposition.

Proposition 8

Suppose we have $V \subset \mathbb{A}_k^n$, $W \subset \mathbb{A}_k^m$, and we write $f = (f_1, \ldots, f_m)$, then

$$f : V \to W \text{ is polynomial map } \iff \forall j, f_j = Y_j \circ f \in k[W]$$

by given Y_j j-th coordinate function.

- Given polynomial map $f : V \to W$, now we define map

$$f^* : k[W] \to k[V]$$
$$g \mapsto g \circ f$$

- Then f^* is ring homomorphism, and in fact, it is k-algebra homomorphism.
- Note that it went backwards: from coordinate ring of the target to that of the source.
- Now with that, we propose a claim

Proposition 9

Every k-algebra homomorphism $\Phi\colon k[W] \to k[V]$ is of the form $\Phi = f^*$ for a *uniquely defined* polynomial map $f\colon V \to W$.

- Before defining affine variety, we want the following claim.

Proposition 10

Given $V \subset \mathbb{A}_k^n$, we have

$$V \text{ is irreducible} \iff I(V) \text{ prime ideal} \iff k[V] \text{ is integral domain}$$

Sketch proof. We shall prove the contrapositive X reducible $\leftrightarrow$ $I(X)$ is not prime. If X is reducible, then we have $X = X_1 \cup X_2$ and choose $f_1 = I(X_1)\backslash I(X)$ and $f_2 = I(X_2)\backslash I(X)$, hence $f_1, f_2 \notin I(X)$ but $f_1 f_2 \in I(X)$ since it vanishes there. If the ideal is not prime, then $\exists f_1, f_2 \notin I(X)$ but $f_1 f_2 \in I(X)$. Construct $I_1 = (I(X), f_1)$ and $X_1 = V(I_1)$, while samewise with $X_2 = V(I_2)$, then $X = X_1 \cup X_2$. The second iff is obvious since $k[V] = k[X_1, \ldots, X_n]/I(V)$.

- We want V be irreducible, so that we can use $k[V]$ to define a quotient field, which will be shown later.

Affine Varieties

Definition 11 (Affine varieties)

An *affine variety* over a field k is an irreducible algebraic subset $V \subset \mathbb{A}^n_k$, equipped with its coordinate ring $k[V]$. Equivalently, it is a set V together with a ring $k[V]$ of k-valued functions $f : V \to k$ such that:

1. $k[V]$ is a finitely generated k-algebra, and
2. for some choice $x_1, \ldots, x_n$ of generators of $k[V]$ over k, the map

$$V \to \mathbb{A}^n_k, \qquad P \mapsto \big(x_1(P), \ldots, x_n(P)\big)$$

embeds V as an irreducible algebraic set.

Morphism between Affine Varieties

- Now we wish to know the mapping between affine varieties.
- Intuitively, we would use usual mapping consists of polynomial function, as previous shown.
- However, it is too restrictive as we can have rational function.
- For example, suppose we have Pythogoras equation $X^2 + Y^2 = Z^2$ and we wish to transform it into circle by $x = X/Z$ and $y = Y/Z$, so we have $x^2 + y^2 = 1$ so finally we can have parametrisation of $\lambda = \frac{x}{1-y}$, hence $x = \frac{2\lambda}{\lambda^2+1}$ and $y = \frac{\lambda^2-1}{\lambda^2+1}$
- And notice that using the mapping of $f(x) = 1/x$ is not polynomial, hence we need to define rational mapping.

Rational Function

- Given V is irreducible and $k[V]$ domain integral, then define

$$k(V) := \text{Quot}(k[V])$$

  as $\text{Quot}(R)$ is quotient field of domain integral R.
- By this, we can write $f \in k(V)$ as $f = g/h$ for $h \neq 0$ (in the sense of zero polynomial).
- But notice f is not well defined if $\exists P \in V$ such that $h(P) = 0$.
- Hence we define as following

Definition 12

Let $f \in k(V)$ and $P \in V$, define f is *regular* at P if $h(P) \neq 0$ with $f = g/h, g, h \in k[V]$, and define *domain of definition* of f by

$$\text{dom}\, f := \{P \in V \mid f \text{ is regular at } P\}$$

and define *local ring* of V at P as

$$\mathcal{O}_{V,P} := \{f \in k(V) \mid f \text{ is regular at } P\}$$

- We seek the characteristics of the $\operatorname{dom} f$, define $D_f := \{h \in k[V] \mid hf \in k[V]\}$, then

$$V \backslash \operatorname{dom} f = \{P \in V \mid h(P) = 0\} = V(D_f)$$

then $\operatorname{dom} f$ open in Zariski Topology.

References

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